AA Part 145 Compliance Analysis Syste

Architecture & Workflow Guide

Facility: Skysource Aviation, LLC

Certificate Number: HSYR817E

Primary Regulation: 14 CFR Part 145 - Repair Stations

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Executive Summary

This document describes an automated compliance analysis system designed to evaluate FAA Part 145 repair station documentation against regulatory requirements. The system combines indexed regulatory documents, specialized research capabilities, and intelligent agents to perform comprehensive audits that would traditionally require significant manual effort.

Key Capabilities

- **Automated Regulatory Research** — Searches across 110+ indexed FAA documents including regulations, advisory circulars, and inspector guidance
- **DCT-Based Evaluation** — Applies the same Data Collection Tools (DCTs) used by FAA inspectors, covering 512 evaluation questions across 18 forms
- **Dual-Track Recommendations** — Generates both minimum compliance and best practice options for every finding
- **Continuous Learning** — Captures insights from each audit to improve future evaluations
- **Professional Reporting** — Produces structured findings in multiple formats (JSON, Markdown, Word)

System at a Glance

18	512	110+	100%
DCT FORMS	QUESTIONS	DOCUMENTS INDEXED	COMPLETE

Component	Status
DCT Forms Evaluated	18 OF 18 (100%)
Total Questions Processed	512
Documents Indexed	110+ markdown files
Overall Facility Status	SATISFACTORY

1. Introduction: The Challenge of Compliance Auditing

The Traditional Approach

FAA Part 145 repair stations must demonstrate compliance with dozens of regulatory requirements. Traditionally, this involves:

- Manually reviewing repair station manuals against CFR requirements
- Cross-referencing multiple Advisory Circulars for acceptable methods
- Applying FAA Order 8900.1 inspector guidance to evaluate procedures
- Documenting findings for each of the 512 questions across 18 DCT forms
- Generating corrective action recommendations with regulatory citations

This process is time-intensive, requires deep regulatory expertise, and is prone to inconsistency between auditors.

The Automated Solution

This system transforms compliance auditing by:

1. **Indexing all regulatory sources** into a searchable knowledge base
2. **Automating research** across CFR, Advisory Circulars, and FAA Orders
3. **Applying DCT evaluation criteria** question-by-question with evidence documentation
4. **Generating recommendations** with proper regulatory citations
5. **Learning from each audit** to improve future evaluations

Benefits for Repair Stations

Benefit	Description
Consistency	Same evaluation criteria applied uniformly across all documents
Traceability	Every finding linked to specific regulatory requirements
Efficiency	Automated research eliminates manual document searching
Completeness	All 512 DCT questions evaluated systematically
Actionable Output	Two implementation paths for every finding

2. System Overview

The system is built on five interconnected pillars that work together to perform comprehensive compliance analysis.

The Five Pillars

SYSTEM ARCHITECTURE OVERVIEW

```
┌──────────────────────────────────────────────────────────────────────────┐
│                    FAA COMPLIANCE ANALYSIS SYSTEM                        │
├──────────────────────────────────────────────────────────────────────────┤
│                                                                          │
│   ┌──────────┐    ┌──────────┐    ┌──────────┐    ┌──────────┐    ┌──────────┐ │
│   | SOURCE  |    |  INDEX  |    |RESEARCH |    |EVALUATION|    |REPORTING| │
│   |  DOCS   |──►| SYSTEM  |──►| SKILLS  |──►|  AGENTS  |──►| OUTPUT  | │
│   └──────────┘    └──────────┘    └──────────┘    └──────────┘    └──────────┘ │
│                                                                          │
│   14 CFR        Searchable    Find & Cross   Apply DCT     Generate    │
│   Order 8900    Knowledge     Reference      Criteria      Reports     │
│   Adv Circulars Base          Requirements   Question      with        │
│   DCT Forms                                  by Question   Citations   │
│   RS Documents                                                         │
│                                                                          │
└──────────────────────────────────────────────────────────────────────────┘
```

Document Authority Hierarchy

The system understands the relationship between different regulatory documents. This hierarchy determines how requirements are interpreted and applied:

REGULATORY AUTHORITY PYRAMID

```
┌──────────────────────┐
|    14 CFR Part 145  |
|    (REGULATION)     |
|   Primary Authority |
└──────────┬───────────┘
           |
           | Interpreted by
           ▼
┌──────────────────────┐
|  FAA Order 8900.1   |
|(INSPECTOR GUIDANCE) |
| How FAA Evaluates   |
└──────────┬───────────┘
           |
           | Supported by
           ▼
┌──────────────────────┐
| Advisory Circulars  |
|    (GUIDANCE)       |
| Acceptable Methods  |
└──────────┬───────────┘
           |
           | Implemented in
           ▼
┌──────────────────────┐
| Repair Station Docs |
|  (COMPLIANCE)       |
| Company Procedures  |
└──────────┬───────────┘
           |
           | Evaluated using
           ▼
┌──────────────────────┐
|     DCT Forms       |
|   (ASSESSMENT)      |
| 512 Questions       |
└──────────────────────┘
```

Key Statistics

Category	Count	Description
CFR Sections	32	14 CFR Part 145 sections indexed
Advisory Circulars	60	FAA guidance documents cataloged
Order 8900.1 Sections	30	Inspector guidance chapters
DCT Forms	18	Evaluation instruments
DCT Questions	512	Individual evaluation criteria
Repair Station Documents	2	RSTM + RSQC manuals
Total Indexed Files	110+	Markdown format for efficiency

3. Source Documents: The Regulatory Foundation

The system draws from five categories of source documents, each serving a specific purpose in the compliance evaluation process.

DOCUMENT TYPES & RELATIONSHIPS

```
┌──────────────────────────────────────────────────────────────────────────┐
│                          SOURCE DOCUMENTS                                │
├──────────────────────────────────────────────────────────────────────────┤
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  │                        14 CFR PART 145                           │    │
│  │  "The Law" - Mandatory requirements for repair stations          │    │
│  │  Format: XML  |  Sections: 145.1 through 145.223                 |    |
│  └──────────────────────────────────────────────────────────────────┘    │
│                                    |                                      |
│                                    ▼                                      |
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  │                      FAA ORDER 8900.1                            │    │
│  │  Inspector guidance - How FAA evaluates compliance               |    |
│  │  Format: 30 Markdown files  |  Volume 6: Repair Stations         |    |
│  └──────────────────────────────────────────────────────────────────┘    │
│                                    |                                      |
│                                    ▼                                      |
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  |                     ADVISORY CIRCULARS                           |    |
│  |  Acceptable methods - Not the only way, but a proven way         |    |
│  |  Format: 60 Markdown files  |  Key: AC 145-10, AC 43-9D          |    |
│  └──────────────────────────────────────────────────────────────────┘    │
│                                    |                                      |
│                                    ▼                                      |
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  |                     DCT FORMS (18 TOTAL)                         |    |
│  |  Assessment criteria - Official FAA evaluation instruments       |    |
│  |  Format: 18 Markdown files  |  Questions: 512 total              |    |
│  └──────────────────────────────────────────────────────────────────┘    │
│                                    |                                      |
│                                    ▼                                      |
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  |                   REPAIR STATION DOCUMENTS                       |    |
│  |  Compliance evidence - Company's procedures and policies         |    |
│  |  RSTM 2025: Training Manual  |  RSQC Rev 22: Quality Manual      |    |
│  └──────────────────────────────────────────────────────────────────┘    │
└──────────────────────────────────────────────────────────────────────────┘
```

Document Details

14 CFR Part 145 - The Regulation

The Code of Federal Regulations establishes mandatory requirements. Key sections include:

Section	Topic	Compliance Impact
145.103	Housing and facilities	Physical requirements
145.109	Equipment, materials, and data	Resource requirements
145.163	Training requirements	Personnel qualification
145.207	Repair station manual	Documentation requirements
145.211	Quality control system	Process requirements
145.219	Recordkeeping	Record retention

Advisory Circulars - Acceptable Methods

> **Important:** Advisory Circulars describe AN acceptable method, not THE ONLY method. Repair stations may use alternative approaches that achieve equivalent compliance.

Key ACs for Part 145:

- **AC 145-10** — Repair station training programs
- **AC 43-9D** — Maintenance records
- **AC 145-5** — Quality control systems

Why Markdown Format?

All source documents are converted to markdown format for:

- **Faster processing** — No PDF parsing required
- **Direct text access** — Content immediately searchable
- **Lower resource usage** — Smaller file sizes
- **Better accuracy** — No OCR errors

4. The Index System: Organized Knowledge

The index system transforms raw documents into a searchable, cross-referenced knowledge base.

INDEX ARCHITECTURE

```
┌──────────────────────────────────────────────────────────────────────────┐
│                          INDEX SYSTEM                                    │
├──────────────────────────────────────────────────────────────────────────┤
│                      ┌──────────────────────┐                            │
│                      │   MASTER INDEX       │                            │
│                      │  master-index.json   │                            │
│                      │  Central Registry    │                            │
│                      └──────────┬───────────┘                            │
│                                 │                                         │
│            ┌────────────────────┼────────────────────┐                   │
│            │                    │                    │                   │
│            ▼                    ▼                    ▼                   │
│   ┌────────────────┐  ┌────────────────┐  ┌────────────────┐            │
│   │   CFR INDEX    │  │   AC INDEX     │  │  ORDER INDEX   │            │
│   │  part-145.md   │  │  ac-index.md   │  │  order-index.md │           │
│   │  32 sections   │  │  60 documents  │  │  30 sections   │            │
│   └────────────────┘  └────────────────┘  └────────────────┘            │
│                                                                           │
│            ┌────────────────────┼────────────────────┐                   │
│            │                    │                    │                   │
│            ▼                    ▼                    ▼                   │
│   ┌────────────────┐  ┌────────────────┐  ┌────────────────┐            │
│   │   DCT INDEX    │  │   RS DOC INDEX  │  │ CROSS-REFERENCES│           │
│   │  18 forms      │  │  RSTM + RSQC   │  │ Document links  │            │
│   │  512 questions │  │  2 documents   │  │                 │            │
│   └────────────────┘  └────────────────┘  └────────────────┘            │
│                                                                           │
│                      ┌──────────────────────┐                            │
│                      │ ORCHESTRATOR STATE   │                            │
│                      │ Progress tracking    │                            │
│                      │ Results storage      │                            │
│                      └──────────────────────┘                            │
└──────────────────────────────────────────────────────────────────────────┘
```

How Cross-References Work

The system maintains relationships between documents so that searching for one topic reveals all related guidance:

> **Example: Search for "training requirements"**
>
> - **14 CFR 145.163** — The regulatory requirement

- **FAA Order 8900.1 Vol 6, Ch 9, Sec 13** — How FAA evaluates
- **AC 145-10 Section 5** — Acceptable program elements
- **DCT EP_4_3_6 Questions 1-15** — Specific evaluation items
- **RSTM 2025 Section 4.2** — Company's implementation

The DCT Question Index

Each of the 512 DCT questions is indexed with comprehensive metadata:

Field	Description	Example
Question ID	Unique identifier	EP_4_3_6.Q01
QID	FAA tracking number	00011890
Question Text	The evaluation question	"Does the training program address...?"
CFR Reference	Regulatory basis	145.163(a)
Scoping Attribute	Applicability conditions	SMSVP facilities only
Acceptable Responses	Valid answer options	Satisfactory, Unsatisfactory, N/A, N/O

5. The Skill Framework: Specialized Capabilities

The system uses 27 specialized "skills" organized into five categories. Each skill performs a specific function in the compliance analysis workflow.

SKILL CATEGORIES OVERVIEW (27 SKILLS)

```
┌──────────────────────────────────────────────────────────────────────────┐
│                           SKILL FRAMEWORK                                │
├──────────────────────────────────────────────────────────────────────────┤
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  │               INDEX & NAVIGATION SKILLS (3)                      │    │
│  │  /index-source    Parse and index FAA documents                  │    │
│  │  /search-index    Search indexed content by keyword/CFR          │    │
│  │  /navigate-index  Browse index structure, check coverage         │    │
│  └──────────────────────────────────────────────────────────────────┘    │
│                                    |                                      |
│                                    ▼                                      |
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  │                   RESEARCH SKILLS (5)                            │    │
│  │  /research-cfr      Look up 14 CFR sections with analysis        │    │
│  │  /research-ac       Find Advisory Circular guidance              │    │
│  │  /research-8900     Research FAA Order 8900.1 criteria           │    │
│  │  /research-part145  Combined multi-source research               │    │
│  │  /research-agent    Autonomous coordinated research              │    │
│  └──────────────────────────────────────────────────────────────────┘    │
│                                    |                                      |
│                                    ▼                                      |
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  │                  EVALUATION SKILLS (5)                           │    │
│  │  /classify-regulatory  Determine if item is required/optional    │    │
│  │  /audit-dct           Apply DCT criteria to documents            │    │
│  │  /recommend-action    Generate corrective actions                │    │
│  │  /evaluate-simplicity Rate implementation effort                 │    │
│  │  /evaluate-cost       Estimate financial impact                  │    │
│  └──────────────────────────────────────────────────────────────────┘    │
│                                    |                                      |
│                                    ▼                                      |
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  │                    REPORT SKILLS (2)                             │    │
│  │  /generate-report   Compile findings into formatted reports      │    │
│  │  /run-audit         Execute complete audit workflow              │    │
│  └──────────────────────────────────────────────────────────────────┘    │
│                                    |                                      |
│                                    ▼                                      |
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  │                     AGENT SKILLS (4)                             │    │
│  │  /pmi-agent      Autonomous FAA inspector role                   │    │
│  │  /am-agent       Autonomous management advisor role              │    │
│  │  /full-audit     Complete PMI + AM + Report pipeline             │    │
│  │  /orchestrator   Master controller for all 18 DCTs               │    │
│  └──────────────────────────────────────────────────────────────────┘    │
└──────────────────────────────────────────────────────────────────────────┘
```

How Skills Chain Together

Skills work together in sequences. Here's how a single evaluation question flows through the skill chain:

SKILL WORKFLOW CHAIN

```
                        ┌──────────────────┐
                        | DCT Question:    |
                        | "Does training   |
                        |  program exist?"|
                        └──────────────────┘
                                 |
              ┌──────────────────┼──────────────────┐
              |                  |                  |
              ▼                  ▼                  ▼
      ┌──────────────────┐ ┌──────────────────┐ ┌──────────────────┐
      | /research-cfr    | | /research-ac     | | /research-8900   |
      | Find: 145.163    | | Find: AC 145-10  | | Find: Vol 6 Ch 9|
      └──────────────────┘ └──────────────────┘ └──────────────────┘
              |                  |                  |
              └──────────────────┼──────────────────┘
                                 |
                                 ▼
                     ┌──────────────────────┐
                     |    /audit-dct        |
                     | Evaluate document    |
                     | against criteria     |
                     └──────────────────────┘
                                |
                                ▼
                     ┌──────────────────────┐
                     |/classify-regulatory |
                     | Is this REQUIRED     |
                     | or RECOMMENDED?      |
                     └──────────────────────┘
                                |
              ┌─────────────────┼─────────────────┐
              |                                   |
              ▼                                   ▼
     ┌──────────────────┐                 ┌──────────────────┐
     | UNSATISFACTORY   |                 |  SATISFACTORY    |
     | /recommend-action|                 | Document finding|
     | Generate fix     |                 | Move to next Q   |
     └──────────────────┘                 └──────────────────┘
              |
     ┌────────┼────────┐
     ▼                 ▼
┌──────────────┐  ┌──────────────┐
|/evaluate-    | |/evaluate-    |
| simplicity   | | cost         |
| Effort: LOW  | | Cost: $2,400|
└──────────────┘  └──────────────┘
```

6. The Agent System: Intelligent Automation

The system employs three intelligent agents, each playing a specific role in the compliance evaluation process. These agents work autonomously but hand off work to each other in a coordinated workflow.

THREE-AGENT ARCHITECTURE

```
┌──────────────────────────────────────────────────────────────────────────┐
│                          AGENT WORKFLOW                                  │
├──────────────────────────────────────────────────────────────────────────┤
│                          ┌──────────────────┐                            │
│                          │  USER REQUEST    │                            │
│                          │ "Audit RSTM"     │                            │
│                          └──────────┬───────┘                            │
│                                     │                                     │
│                                     ▼                                     │
│  ┌──────────────────────────────────────────────────────────────────┐   │
│  │                       RESEARCH AGENT                             │   │
│  │                   "The Subject Matter Expert"                    │   │
│  │                                                                  │   │
│  │  Role: Find all relevant regulatory requirements                 │   │
│  │  • Search 14 CFR Part 145 for applicable sections                │   │
│  │  • Find related Advisory Circulars                               │   │
│  │  • Locate FAA Order 8900.1 evaluation guidance                   │   │
│  │                                                                  │   │
│  │  Output: Regulatory context for each topic                       │   │
│  └──────────────────────────────────────────────────────────────────┘   │
│                                     │                                     │
│                                     ▼                                     │
│  ┌──────────────────────────────────────────────────────────────────┐   │
│  │                         PMI AGENT                                │   │
│  │               "The Primary Maintenance Inspector"                │   │
│  │                                                                  │   │
│  │  Role: Evaluate documents using FAA DCT criteria                 │   │
│  │  • Load applicable DCT form(s)                                   │   │
│  │  • For each of 512 questions: Check scoping, search document,    │   │
│  │    compare against acceptable responses, determine status        │   │
│  │  • Classify each finding (REQUIRED vs RECOMMENDED)               │   │
│  │                                                                  │   │
│  │  Output: audit-[DCT]-[Document].json with all findings           │   │
│  └──────────────────────────────────────────────────────────────────┘   │
│                                     │                                     │
│                                     ▼                                     │
│  ┌──────────────────────────────────────────────────────────────────┐   │
│  │                         AM AGENT                                 │   │
│  │                 "The Accountable Manager Advisor"                │   │
│  │                                                                  │   │
│  │  Role: Generate corrective actions and implementation plans      │   │
│  │  • Review all PMI findings                                       │   │
│  │  • For each UNSATISFACTORY: Generate MINIMUM COMPLIANCE option   │   │
│  │    and BEST PRACTICE option                                      │   │
│  │  • Estimate effort and cost, prioritize by safety impact         │   │
│  │                                                                  │   │
│  │  Output: corrective-actions-[Date].json with action plans        │   │
│  └──────────────────────────────────────────────────────────────────┘   │
│                                     │                                     │
│                                     ▼                                     │
│                       ┌──────────────────────┐                           │
│                       │    FINAL REPORT      │                           │
│                       │ All findings + plans │                           │
│                       │ in PDF-ready format  │                           │
```

```
│                          └──────────────────────┘                                  │
└────────────────────────────────────────────────────────────────────────────────────┘
```

The Dual-Recommendation Model

For every finding, the AM Agent generates two implementation options:

Minimum Compliance

Goal: Meet regulatory requirement

- Simplest path to compliance
- Lowest cost and effort
- Satisfies FAA requirements

Example: Add recurrent training date field to existing personnel records

Best Practice

Goal: Exceed regulatory requirement

- Industry-leading approach
- Higher investment
- Sustained operational excellence

Example: Implement competency-based training tracking with LMS integration

7. The Complete Audit Workflow

This section describes the end-to-end flow from initial audit request through final report delivery.

SEVEN-STEP AUDIT PROCESS

```
┌──────────────────────────────────────────────────────────────────────────┐
│                    END-TO-END AUDIT WORKFLOW                             │
├──────────────────────────────────────────────────────────────────────────┤
│                                                                          │
│  STEP 1                        STEP 2                                    │
│  ┌──────────────────────┐      ┌──────────────────────┐                  │
│  │ REQUEST RECEIVED     │      │ DOCUMENT LOADING     │                  │
│  │ "Audit RSTM against  │──────►│ • Load RSTM content  │                  │
│  │  Part 145"           │      │ • Load DCT forms     │                  │
│  │ Determine:           │      │ • Load CFR sections  │                  │
│  │ • Applicable DCTs    │      │ • Load AC guidance   │                  │
│  │ • Scoping rules      │      │ • Check learnings    │                  │
│  └──────────────────────┘      └──────────────────────┘                  │
│                                            │                              │
│                                            ▼                              │
│  STEP 3                        STEP 4                                    │
│  ┌──────────────────────┐      ┌──────────────────────┐                  │
│  │ QUESTION-BY-QUESTION │      │ FINDING              │                  │
│  │ EVALUATION           │      │ CLASSIFICATION       │                  │
│  │ For each of 512 Qs:  │──────►│ For each finding:    │                  │
│  │ • Check scoping      │      │ • Is it REQUIRED?    │                  │
│  │ • Search document    │      │ • Is it RECOMMENDED? │                  │
│  │ • Compare to         │      │ • What CFR applies?  │                  │
│  │   acceptable resp.   │      │ • What priority?     │                  │
│  │ • Determine status   │      │                      │                  │
│  └──────────────────────┘      └──────────────────────┘                  │
│                                            │                              │
│                                            ▼                              │
│  STEP 5                        STEP 6                                    │
│  ┌──────────────────────┐      ┌──────────────────────┐                  │
│  │ CORRECTIVE ACTION    │      │ REPORT               │                  │
│  │ GENERATION           │      │ COMPILATION          │                  │
│  │ For UNSATISFACTORY:  │──────►│ Compile:             │                  │
│  │ • Research basis     │      │ • Executive summary  │                  │
│  │ • Generate minimum   │      │ • All findings       │                  │
│  │   compliance option  │      │ • All corrective     │                  │
│  │ • Generate best      │      │   actions            │                  │
│  │   practice option    │      │ • Implementation     │                  │
│  │ • Estimate effort    │      │   timeline           │                  │
│  └──────────────────────┘      └──────────────────────┘                  │
│                                            │                              │
│                                            ▼                              │
│  STEP 7                                                                   │
│  ┌──────────────────────────────────────────────────────────────────┐    │
│  │                    LEARNING CAPTURE                              │    │
│  │ Extract insights: COMPLIANCE_GAP, BEST_PRACTICE,                 │    │
│  │ DOCUMENTATION_PATTERN, RECURRING_ISSUE                           │    │
│  │ Store in learnings database for future audits                    │    │
│  └──────────────────────────────────────────────────────────────────┘    │
│                                                                          │
└──────────────────────────────────────────────────────────────────────────┘
```

Step Details

Step 3: Question-by-Question Evaluation

For each of the 512 DCT questions, the system performs:

1. **Scoping Check** — Does this question apply to this facility?
2. **Document Search** — Find relevant content in the target document
3. **Comparison** — Compare document content to DCT acceptable responses
4. **Determination** — Assign status (SATISFACTORY, UNSATISFACTORY, N/A, N/O)

Scoping Logic

Some DCT questions only apply to certain facility types:

Facility Attribute	Questions Affected	Action
Non-SMSVP	113 questions	Mark as N/A
No non-cert contractors	21 questions	Mark as N/A
No TCCA supplement	27 questions	Mark as N/A

Finding Classification

Language in CFR	Classification	Action Required
"shall" or "must"	REQUIRED	Correction mandatory
"should"	RECOMMENDED	Correction encouraged
"may"	OPTIONAL	No action needed

8. The Learning System: Continuous Improvement

The system captures insights from every audit and applies them to future evaluations. This creates a virtuous cycle of continuous improvement.

LEARNING FEEDBACK LOOP

```
┌──────────────────────────────────────────────────────────────────────────┐
│                       LEARNING FEEDBACK LOOP                             │
├──────────────────────────────────────────────────────────────────────────┤
│                     ┌──────────────────────┐                             │
│                     │    AUDIT EXECUTION   │                             │
│                     │  PMI Agent evaluates│                             │
│                     │  512 questions       │                             │
│                     └──────────┬───────────┘                             │
│                                │                                          │
│                                ▼                                          │
│   ┌──────────────────────────────────────────────────────────────────┐   │
│   │                    LEARNING CAPTURE                              │   │
│   │  • Extract patterns from UNSATISFACTORY findings                 │   │
│   │  • Note effective documentation approaches                       │   │
│   │  • Record interpretation decisions                               │   │
│   │  • Check for recurring issues (3+ occurrences)                   │   │
│   └──────────────────────────────────────────────────────────────────┘   │
│                                │                                          │
│                                ▼                                          │
│                     ┌──────────────────────┐                             │
│                     │   LEARNINGS INDEX    │◄────────────────┐           │
│                     │  Indexed by:         │                 │           │
│                     │  • CFR section       │                 │           │
│                     │  • Pattern type      │                 │           │
│                     │  • Document type     │                 │           │
│                     └──────────┬───────────┘                 │           │
│                                │                             │           │
│                                ▼                             │           │
│   ┌──────────────────────────────────────────────────────────────────┐   │
│   │                   LEARNING APPLICATION                           │   │
│   │  Before evaluating each DCT question:                            │   │
│   │  • Query learnings index by CFR section                          │   │
│   │  • Flag COMPLIANCE_GAP items for extra scrutiny                  │   │
│   │  • Reference BEST_PRACTICE recommendations                       │   │
│   └──────────────────────────────────────────────────────────────────┘   │
│                                │                                          │
│                                ▼                                          │
│                     ┌──────────────────────┐                             │
│                     │   IMPROVED AUDITS    │──────────────────┘          │
│                     │  • More consistent   │                             │
│                     │  • Better evidence   │                             │
│                     │  • Faster research   │                             │
│                     └──────────────────────┘                             │
└──────────────────────────────────────────────────────────────────────────┘
```

Learning Types

Type	Trigger	Priority	Example
COMPLIANCE_GAP	UNSATISFACTORY finding	HIGH	"Training records missing recurrent dates"
BEST_PRACTICE	Recommended improvement	LOW	"Use electronic signatures for signoffs"
DOCUMENTATION_PATTERN	Effective evidence	MEDIUM	"Reference QC manual section numbers"
PROCESS_INSIGHT	Audit observation	MEDIUM	"CORRIDOR tracks maintenance history"
RECURRING_ISSUE	3+ similar findings	HIGH	"Contractor oversight frequently incomplete"

Current Learning Statistics

Metric	Value
Total Learnings Captured	25
Best Practices	12
Documentation Patterns	11
Observable Patterns	2
Compliance Gaps	0 (reflects SATISFACTORY status)
Recurring Issues	0

9. Outputs & Deliverables

The system produces findings in multiple formats to serve different purposes.

Output File Types

Format	Purpose	Use Case
JSON (.json)	Machine-readable structured data	Import into compliance management systems
Markdown (.md)	Human-readable formatted text	Review, collaboration, version control
Word (.docx)	Professional delivery document	Management review, regulatory submission

File Naming Convention

Standard Format: `audit-{DCT_ID}-{Title-With-Dashes}.{extension}`

Examples:

- audit-EP_4_3_6-Maintenance-Process-Surveillance.docx
- audit-EP_1_4_1-Personnel-Records-Initial-Cert.json
- audit-EP_4_4_6-Record-Systems-Surveillance.md

Output Structure for Each DCT

File	Content	Size (typical)
audit-EP_X_X_X-findings.json	Complete structured data	15-25 KB
audit-EP_X_X_X-findings.md	Formatted markdown	10-20 KB
audit-EP_X_X_X-[Title].docx	Professional Word doc	38-62 KB

Final Report Summary

Category	Surveillance	Initial Cert	Total
DCT Forms	11	7	18
Questions	320	192	512
Reports Generated	11	7	18

10. Summary & Key Takeaways

What Makes This System Effective

Capability	Value
Comprehensive Coverage	All 18 DCT forms, 512 questions evaluated systematically
Regulatory Traceability	Every finding linked to specific CFR sections
Dual Recommendations	Both minimum compliance and best practice options
Continuous Learning	Insights from each audit improve future evaluations
Professional Output	Multiple formats for different stakeholder needs
Scoping Intelligence	Automatic N/A handling for facility-specific attributes

Current Completion Status

Facility Evaluation Summary: Skysource Aviation (HSYR817E)

18/18	512	100%	PASS
DCT FORMS	QUESTIONS	COMPLETION	STATUS

Results Breakdown

220 Satisfactory | **276** Not Applicable | **16** Not Observable | **0** Unsatisfactory

Overall Facility Status: SATISFACTORY

System Architecture Summary

COMPLETE SYSTEM FLOW

```
SOURCE DOCUMENTS          INDEX SYSTEM          SKILL FRAMEWORK
┌──────────────┐          ┌──────────────┐        ┌──────────────┐
| 14 CFR      |          | Master Index|       | 27 Skills   |
| Order 8900.1|──Parse──▶| Cross-Refs  |──Use─▶| in 5        |
| Adv Circ    |          | DCT Index   |       | Categories  |
| DCT Forms   |          | RS Index    |       |             |
| RS Docs     |          └──────────────┘        └──────────────┘
└──────────────┘                                        |
      |                                                 |
      |                                                 ▼
      |                 AGENT SYSTEM           LEARNING SYSTEM
      |                 ┌──────────────┐        ┌──────────────┐
      |                 | Research    |        | 25 Learnings|
      └──────Evaluate──▶| PMI Agent   |──Feed─▶| Indexed by  |
                        | AM Agent    |        | CFR/Type/Doc|
                        └──────────────┘        └──────────────┘
                              |                       |
                              ▼                       ▼
                        OUTPUT FILES           FUTURE AUDITS
                        ┌──────────────┐        ┌──────────────┐
                        | JSON        |        | Better      |
                        | Markdown    |        | Consistency |
                        | Word        |        | Faster      |
                        | 18 Reports  |        | Smarter     |
                        └──────────────┘        └──────────────┘
```

Appendix A: DCT Form Inventory

Initial Certification DCTs (7 Forms)

Form ID	Title	Questions
EP_1_4_1	Personnel Records	22
EP_1_4_2	Certificate Requirements	18
EP_1_4_3	Manuals	31
EP_1_4_4	Quality Control System	45
EP_1_5_1	Safety Policy	24
EP_1_5_2	Safety Risk Management	28
EP_1_5_4	Safety Promotion	24

Surveillance DCTs (11 Forms)

Form ID	Title	Questions	Primary CFR
EP_4_1_4	Training & Qualification	40	145.163
EP_4_2_6	Technical Data	22	145.109
EP_4_2_9	TCCA Special Conditions	27	145.109
EP_4_3_6	Maintenance Process	40	145.207/209
EP_4_3_7	Work Away from Station	18	145.203
EP_4_4_6	Record Systems	35	145.219
EP_4_5_4	Housing & Facilities	28	145.103
EP_4_5_5	Contract Work (Certificated)	24	145.217
EP_4_5_6	Contract Work (Non-Cert)	21	145.217
EP_4_7_3	Tools & Equipment	32	145.109
EP_4_7_4	Parts & Materials	26	145.109

Appendix B: Glossary

Term	Definition
14 CFR	Title 14 of the Code of Federal Regulations (aviation regulations)
Advisory Circular (AC)	FAA guidance document describing acceptable compliance methods
AM Agent	Accountable Manager Agent - generates corrective action plans
CFR	Code of Federal Regulations
DCT	Data Collection Tool - FAA evaluation instrument
FAA Order 8900.1	Flight Standards Information Management System (inspector guidance)
N/A	Not Applicable - question does not apply to this facility
N/O	Not Observable - cannot evaluate from documentation alone
Part 145	CFR section governing repair station certification
PMI Agent	Primary Maintenance Inspector Agent - evaluates compliance
RSTM	Repair Station Training Manual
RSQC	Repair Station Quality Control Manual
SMSVP	Safety Management System Voluntary Program

Document generated by the FAA Part 145 Compliance Analysis System

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